

CS 160 — Team 4

Sprint 1 Software Engineering Report

Voice Assistant for Older Adults

San Jose State University, CS 160 Software Engineering, Fall 2026

Instructor: Dominic Abucejo

Sprint 1: Week 6 – 7

Table of Contents

1. Git Account Profile Member List
2. Survey Report and Key Feature Decision
3. Project Scope
4. High-Level Architectural Block Diagram
5. Component Diagram
6. Sprint 1 Sequence Diagram
7. Class Diagram
8. Product Backlog
9. Sprint 1 Backlog and Task Assignment
10. Sprint Summary
11. Plans for Next Sprint

1. Git Account Profile Member List

Team Member	GitHub Username
Zohreh Ashtarilarki	ZohrehAshtarilarki
Kunj Shah	kunjcr2
Amarjargal Ayurzana	Akiko0210
Moebius Yang	

Repository URL: https://github.com/kunjcr2/CS160_Project

2. Survey Report and Key Feature Decision

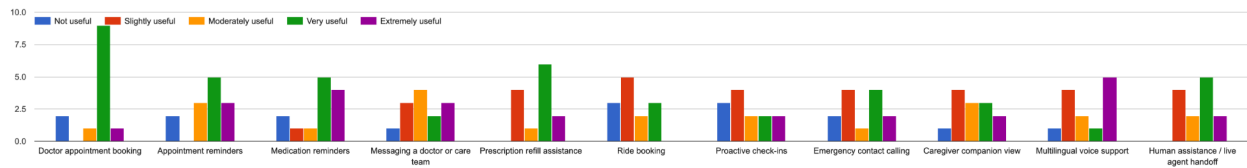
The survey collected 13 responses. All 13 respondents identified themselves as people who help care for or support someone age 60 or older. Therefore, the evidence is useful for caregiver/supporter priorities but does not directly represent older adults answering for themselves.

The team reviewed both the average usefulness ratings and the Top 3 selections from the survey when selecting the seven features for implementation. The final selection also considered the goals and scope of the project. The voice conversational interface and confirmation-before-action behavior are supporting system requirements used across the selected features and are not counted as separate core features.

Selected feature	Average Usefulness	Extremely Useful	Top-3 Votes
Doctor Appointment Booking	3.54/5	10	12
Appointment Reminders	3.54/5	8	9
Medication Reminders	3.62/5	9	8
Prescription Refill Assistance	3.46/5	8	0
Human Assistance / Live Agent Handoff	3.38/5	7	1
Multilingual Voice Support	3.38/5	6	3
Messaging a Doctor or Care Team	3.23/5	5	3

Survey: Feature Usefulness Chart

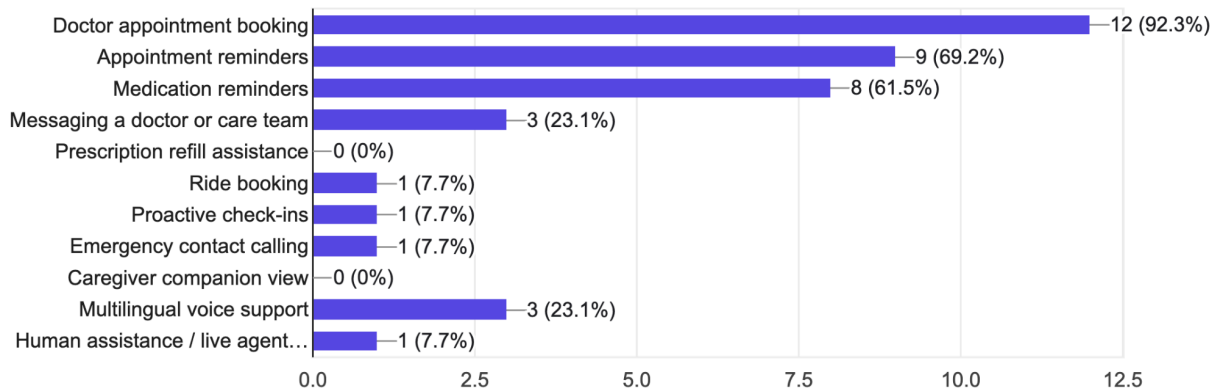
9. How useful would each of the following features be for you or the older adult you support?



Survey: Top-3 Feature Priorities

10. If we could only build THREE of these features, which three should we prioritize?

13 responses



Key interpretation: the survey results show that appointment booking, appointment reminders, and medication reminders were the most useful features to respondents. Multilingual support was also important. Eleven out of 13 respondents rated support for their preferred language as important or extremely important, and all 13 said this feature would make them more likely, or possibly more likely, to use the product. Human assistance was also kept because some users wanted to connect directly with a person, while others preferred having a caregiver contacted if the system kept having trouble understanding them.

3. Project Scope

Based on the survey results, the team selected the following seven core features for implementation:

1. Medication Reminders

Remind users when it is time to take their medications and allow them to confirm that the medication was taken.

2. Doctor Appointment Booking

Allow users to search for available appointments and book a doctor's appointment through the assistant.

3. Appointment Reminders

Notify users about upcoming doctor appointments.

4. Prescription Refill Assistance

Allow users to request prescription refills through the assistant.

5. Multilingual Voice Support

Allow users to interact with the assistant in supported languages other than English.

6. Human Assistance / Live-Agent Handoff

Allow users to request human assistance when the automated assistant cannot successfully complete their request.

7. Messaging a Doctor/Care Team

Allow users to create and send messages to their doctor or care team through the assistant.

3.1. Supporting System Requirements

The seven core features will be supported by a voice-based conversational interface, confirmation before important actions, accessible error handling, and fallback behavior.

During early development, the system may use text or speech transcripts to test conversational workflows before full voice integration is completed.

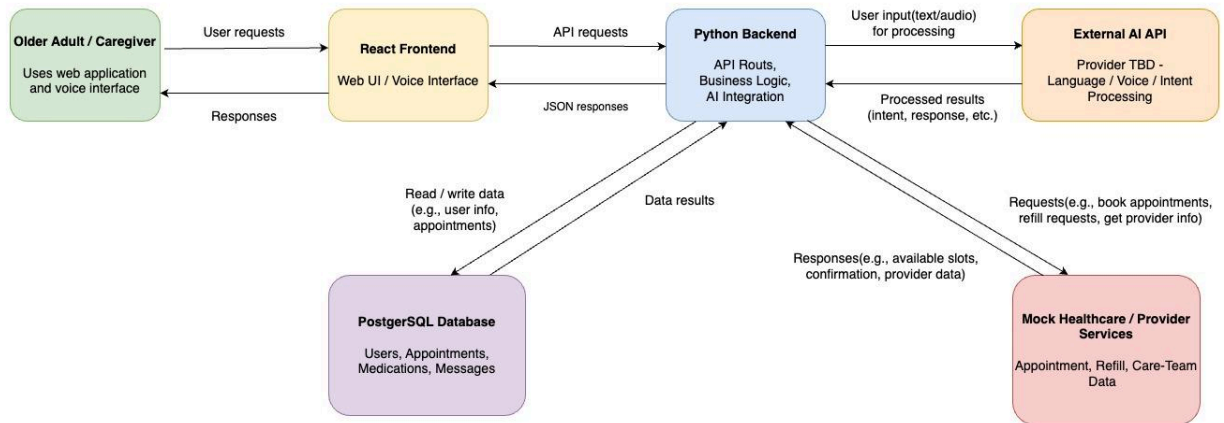
For the prototype, healthcare operations will use mock provider services and seeded demonstration data instead of connecting directly to real hospital systems.

3.2. Sprint 1 Goal

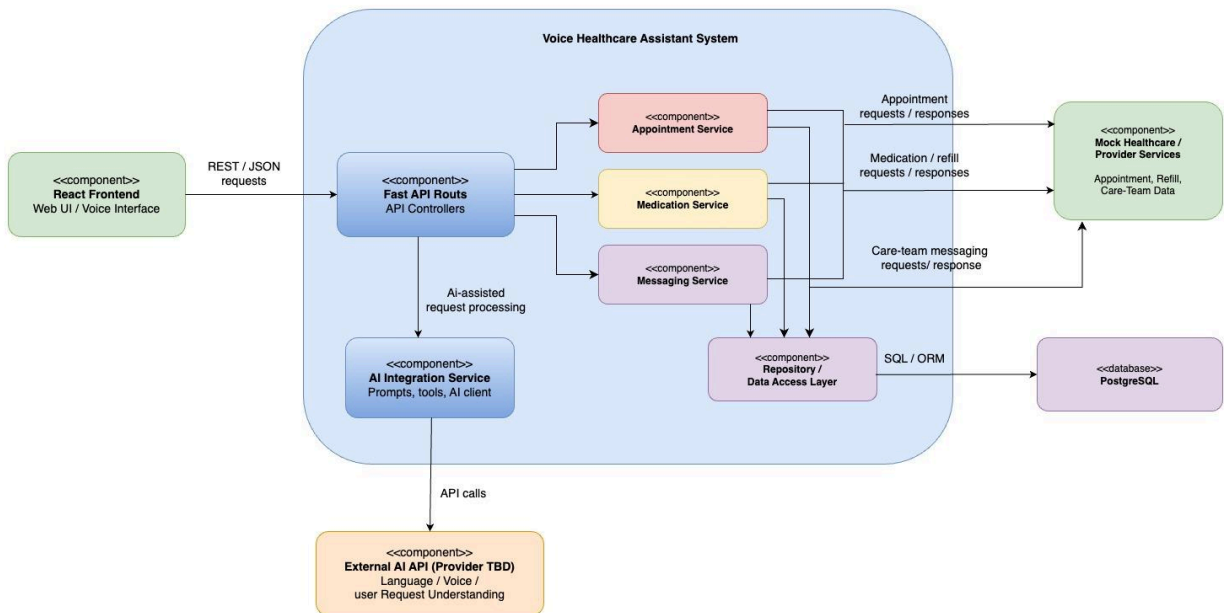
The goal of Sprint 1 is to complete the project planning and specification. The team will finalize the seven core features, create the architecture and design diagrams, prepare the product and sprint backlogs, assign tasks to team members, organize the project documentation, and set up the shared GitHub repository.

4. High-Level Architectural Block Diagram

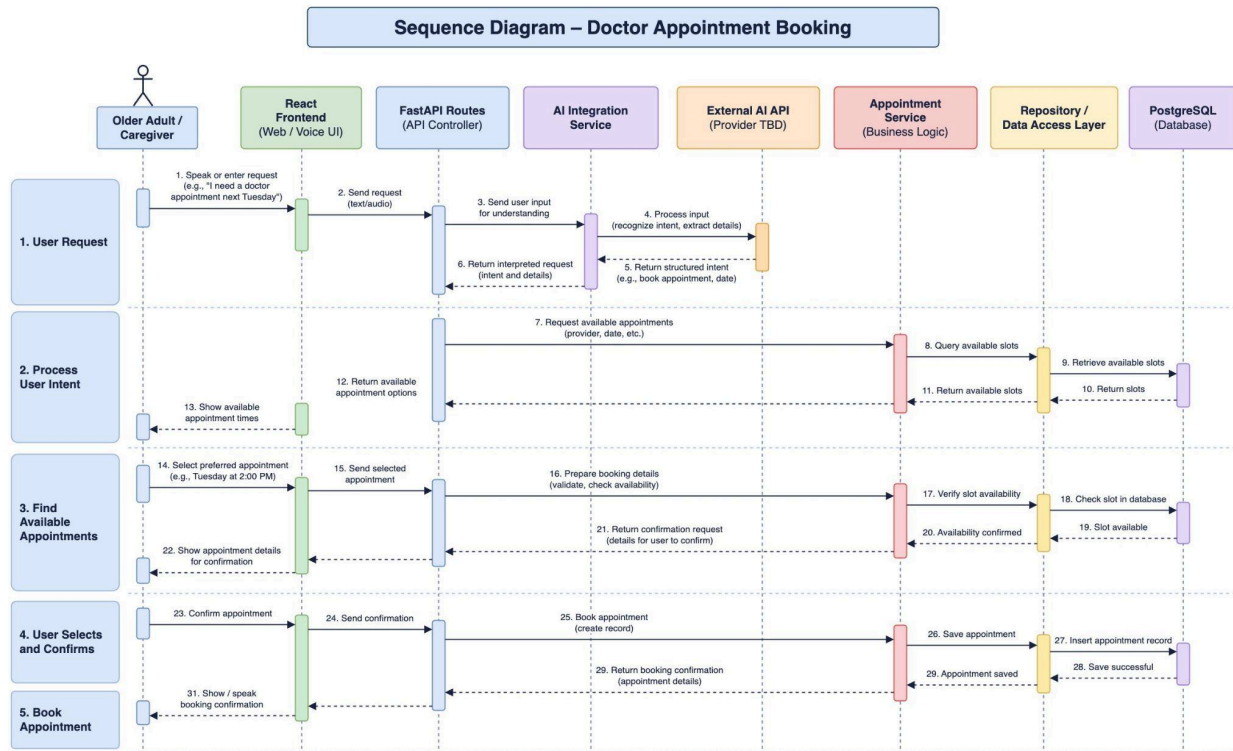
MVC mapping: React serves as the View, and FastAPI routes act as the Controllers. The service layer contains the main business logic, while the domain models and repository layer manage application data and communication with PostgreSQL. The ai/ package handles communication with the external AI API and is part of the backend, not a separate server.



5. Component Diagram



6. Sprint 1 Sequence Diagram



7. Class Diagram

- Major blocker(s): [technical/team/dependency]
- What went well: [specific evidence]
- What to improve next sprint: [specific process/technical change]

11. Plans for the Next Sprint

- Complete any unfinished Sprint 1 documentation or planning tasks.
- Begin implementation of the project based on the finalized specifications and architecture.
- Set up the React frontend, Python/FastAPI backend, and PostgreSQL database.
- Finalize the external AI API/provider that will be used for the project.
- Begin implementing the first selected feature based on team priority and workload.
- Connect the frontend, backend, and database as needed for the first feature.
- Begin testing the initial implementation.